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To help with calculus, here are the most common trigonometric identities grouped by how they are typically used.

1. Power-Reduction Identities

These are essential for integrating squared trigonometric functions, just like the one you just solved.

- $\cos^2 x = \frac{1+\cos(2x)}{2}$
- $\sin^2 x = \frac{1-\cos(2x)}{2}$
- $\tan^2 x = \frac{1-\cos(2x)}{1+\cos(2x)}$

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2. Pythagorean Identities

These are often used to simplify expressions or perform substitutions (like $1 - \sin^2 x = \cos^2 x$).

- $\sin^2 x + \cos^2 x = 1$

Ask anything

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- $\sin^2 x + \cos^2 x = 1$
- $1 + \tan^2 x = \sec^2 x$
- $1 + \cot^2 x = \csc^2 x$

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3. Double-Angle Identities

Useful for simplifying functions before integrating or after using the power-reduction rules.

- $\sin(2x) = 2\sin x \cos x$
- $\cos(2x) = \cos^2 x - \sin^2 x$
  - *Also:* $2\cos^2 x - 1$ *or* $1 - 2\sin^2 x$
- $\tan(2x) = \frac{2\tan x}{1-\tan^2 x}$

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4. Reciprocal & Quotient Identities

The basic building blocks for rewriting integrals.

- $\sec x = \frac{1}{\cos x}$
- $\csc x = \frac{1}{\sin x}$

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5. Product-to-Sum Identities

Used when you need to integrate the product of two different frequencies, like $\int \sin(2x)\cos(3x)\,dx$.

$$\sin A\cos B = \frac{1}{2}[\sin(A-B)+\sin(A+B$$

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- $\cos A\cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$
- $\sin A\sin B = \frac{1}{2}[\cos(A-B)-\cos(A+B)]$

Would you like to practice a problem using the $\sin^2 x$ identity or a **double-angle** one?

AI responses may include mistakes. Learn more

Ask anything

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